

Short Communication

Application of solid-phase extraction to brewed coffee caffeine and organic acid determination by UV/HPLC

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Abstract

A solid-phase extraction (SPE) method was adapted to perform brewed coffee sample clean-up for seven organic acids (acetic, citric, formic, malic, pyruvic, quinic and succinic) and caffeine determination by reversed-phase UV high performance liquid chromatography (RP/UV-HPLC). For all analytes the method showed good precision and linearity and, as an application, 20 brewed coffee samples from the two types of coffee (robusta vs. arabica) were tested. Brewed coffee samples were prepared according to ISO 6668 [1991. Green coffee—preparation of samples for use in sensory analysis] and the results were compared to sensory evaluation obtained from a panel of coffee tasters. Robusta coffee demonstrated the highest content of caffeine. Total acid content varied with coffee type and also with the geographic origin of the green coffee. Roasting conditions also seem to affect final acidity in brewed coffee for both coffee types analysed. © 2007 Elsevier Inc. All rights reserved.

Keywords: Organic acids; Caffeine; Brewed coffee; UV/HPLC; Solid-phase extraction

1. Introduction

Acidity is an important feature in coffee and affects how consumers react to it. The acidity and sourness of coffee brews (together with aroma and bitterness) have always been recognized as important attributes of their sensory quality (Ginz et al., 2000). Marbrouk and Deatherage established in 1956 that the acidity of coffee depends upon the degree of roast (Balzer, 2001). Although malic and citric acids are already present in the green coffee bean, most of the acidity is generated at the beginning of the roasting process. The main acids in green coffee beans are citric, malic, chlorogenic and quinic. During the roasting process the first three acids decrease while quinic acid increases as a result of the degradation of chlorogenic acids (Ginz et al., 2000; Leloup et al., 1995 in Balzer, 2001). Formic and acetic acids yields increase up to the medium

roasting degree and then begin to fall as roasting is continued. According to Balzer (2001), a rapid increase in titratable acidity during roasting was observed from green to medium roast, followed by a smaller decrease as roasting proceeded. The corresponding precursors and formation mechanisms of aliphatic acids in coffee have been shown by Ginz et al. (2000), where the carbohydrate fraction, in particular sucrose, was shown to be the main precursor of formic, acetic, glycolic and lactic acids. Acid contents as a function of organic roast loss for Colombian arabica coffee and Indonesian robusta coffee (Weers et al., 1995 in Balzer, 2001) were determined by capillary electrophoresis (CE) showing several other acids present in brewed coffee such as malic, citric, succinic, phosphoric and quinic. The analysis of acid content of coffee extracts was performed by Verardo et al. (2002) by gas chromatography–mass spectrometry (GC–MS). These authors identified the corresponding butyl and methyl esters of 6 low-molecular weight and 15 high-molecular weight organic acids. Also, according to Balzer (2001), several methods have been applied for organic acid determinations, e.g. GC–MS, CE

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and high performance liquid chromatography (HPLC)/UV, but with green coffee samples or roasted coffee from only one or two geographic origins or types of coffee (robusta or arabica).

The aim of this study was to apply solid-phase extraction (SPE) to brewed coffee in order to determine the organic acid and caffeine content using reversed phase (RP)-UV/HPLC. Organic acid SPE has been applied to other matrixes like fruit juices (Chinnici et al., 2005), honey (Suárez-Luque et al., 2002b) and beer (Belke and Irwin in Alonso Garcia and Simal Gándara, 2004). Suárez-Luque et al. (2002a) report higher recovery rates when using an anion exchange SPE cartridge compared to a reversed-phase (C18) cartridge for organic acids extraction from honey. In our work, an anion exchange cartridge and different extraction parameters were tested to develop an SPE method for carboxylic acid extraction from brewed coffee. The method was tested in several coffee beverages prepared from different types of roasted coffee from 10 different geographic origins and in some cases submitted to distinct roasting conditions (India and Zambia coffee). Although about 100 species (taxa) of the genus *Coffea* have been identified so far, we considered only the two most cultivated and commercialized forms of coffee, robusta coffee (*Coffea canephora*) and arabica coffee (*Coffea arabica*), although depending on the geographic origin of the bean, for each of these two species a range of cultivars can be found as a result of different coffee breeding programs (Van de Vossen, 2001).

In addition, the caffeine content in each brewed coffee sample was determined by (C18)SPE-UV/HPLC. RP-HPLC/UV has already been applied by other authors for caffeine determination in green coffee beans (Barre et al., 1998) and cola beverages (Theodoridis and Manesotis, 2002). SPE has been adapted to extract caffeine from coffee extracts for HPLC determinations according to Theodoridis and Manesotis (2002).

2. Materials and methods

2.1. Samples

Green coffee samples from 10 different geographic origins were selected for this work. The robusta coffees were obtained from India, Viet Nam, Cameroon and Côte d'Ivoire; arabica samples were obtained from Kenya, Honduras, Zambia, Brazil, UR Tanzania and Colombia (Table 1).

The roasting process was performed according to ISO 6668 (1991) (E) (Green coffee: Preparation of samples for use in sensory analysis) at Novadelta, S.A. (Campo Maior, Portugal) at their quality control laboratory. Sensory evaluation was conducted by a trained panel of tasters. After which the brewed coffee samples were collected in 50 mL tubes (Sarstedt, AG and Co.), frozen at -20°C and transported to our laboratory. Samples were kept frozen until SPE.

Table 1

Caffeine content in brewed coffee from different types (robusta vs. arabica) and different geographic origins

Type	Sample identification	Origin	Roast	Caffeine (mg/L)
Robusta	IA	India	Medium	314
	IB			413
	IC			373
	IDm			739
	IDl			762
	IDd	Viet Nam	Light	421
	V		Dark	510
	C1		Medium	327
	C2			394
	Ci			646
Arabica	K	Kenya		190
	H1	Honduras		214
	H2			174
	Z1l	Zambia	Light	303
	Z1d		Dark	322
	Z2		Medium	253
	Z3			456
	B	Brazil		266
	T	UR Tanzania		310
	Co	Colombia		279
	Decaffeinated	—		8

Three Indian brewed coffee samples (all robusta) corresponding to light (150°C , 10 min), medium ($150\text{--}180^{\circ}\text{C}$, 15 min) and dark (220°C , 20 min) roasting conditions were obtained to assess effects of roasting on perceived the acidity and organic acid profiles. The same was carried out with arabica coffee from Zambia.

2.2. Reagents and materials

Analytical standard-grade organic acids were used: quinic, succinic and malic acid (Sigma-Aldrich, Germany), acetic, formic and sodium pyruvate (Merck, USA) and citric acid (Panreac, Spain). Stock solutions were prepared by dissolution of acids in Milli-Q water. The Milli-Q water was purified on a Compact Milli-RO and Milli-Q water system from Millipore (Milford, MA, USA). Working standard solutions were prepared daily by dilution with Milli-Q water.

Samples were filtered through PTFE membrane syringe filters (13 mm, $0.2\text{ }\mu\text{m}$) (VWR International, USA) and SPE was achieved with Strata Sax ($55\text{ }\mu\text{m}$, $70\text{ }\text{\AA}$) 500 mg/3 mL (Phenomenex, CA, USA) for organic acids and with Chromabond 15 mL cartridges filled with 6.0 g of Chromabond C18 Hydra both from Macherey-Nagel (Germany) for caffeine.

HPLC mobile phase for acids determination was prepared with KH_2PO_4 (Riedel-de Haën, Germany) and HPLC-grade methanol (Panreac, Spain). For caffeine determination, the mobile phase was prepared with triethylamine (Riedel-de Haën, Germany) phosphate

buffer and HPLC-grade acetonitrile (Carlo Erba, Italy). Caffeine standard was obtained from Fluka (Germany).

2.3. Apparatus

2.3.1. Organic acid determination

Chromatographic analysis were carried out on a LC-6A Shimadzu liquid chromatograph equipped with a SPD-6A Shimadzu UV/Vis detector. The detector signal was recorded on a Shimadzu C-R3A Chromatopac integrator. The column was a Gemini C18, 5 μ m, 4.6 $\times$ 150 mm with two cartridges of SecurityGuard Gemini C18, 4 $\times$ 3.0 mm (Phenomenex, CA, USA).

2.3.2. Caffeine determination

Caffeine determination was performed in a Merck LaChrom L-7110 liquid chromatograph equipped with a L-7400 Merck Hitachi UV detector. The detector signal was recorded on a D-7500 Merck Hitachi integrator. The column was an Aquasil C18, 5 μ m, 4.6 $\times$ 200 mm with Aquasil C18 SecurityGuard column 10 $\times$ 4/4.6 mm (Thermo Electron, UK).

2.3.3. Solid-phase extraction

SPE was performed in a 12-position Vacuum Manifold (Phenomenex, CA, USA).

2.4. Assay procedure

2.4.1. Solid-phase extraction

2.4.1.1. Organic acids. Brewed coffee samples were diluted (1:2) in 50 mM potassium phosphate buffer, pH 7.0. Anion exchange cartridges were conditioned with 3.0 mL of methanol and 3.0 mL of Milli-Q water. 3.0 mL of diluted sample was passed through the cartridge that was then washed with 2.0 mL of Milli-Q water (pH 7.0).

Acids were eluted with 5 $\times$ 0.5 mL of HCl 1 M. The eluted sample was diluted (1:2) in the HPLC mobile phase, filtered with a 0.2 μ m syringe filter and directly injected in the chromatograph. Analytical procedure for the determination of carboxylic acids in brewed coffee is shown in Fig. 1a.

2.4.1.2. Caffeine. Samples were diluted (1:50) in Milli-Q water (caffeine content of this diluted sample was between 3.5 and 16 mg/L). 6.0 g of C18 Hydra reversed-phase sorbent was packed in 15 mL Chromabond cartridges, washed with Milli-Q water and activated with 10.0 mL of methanol and 10.0 mL of Milli-Q water. A 5.0 mL volume of diluted brewed coffee sample was passed through at a flow-rate of 0.5 mL/min. The cartridge was washed with 5.0 mL of methanol and caffeine was eluted with 2 $\times$ 5.0 mL of methanol. The eluted samples were filtered with 0.2 μ m syringe filters and injected directly in the chromatograph. Analytical procedure for the determination of caffeine in brewed coffee is shown in Fig. 1b.

2.4.2. Chromatographic conditions

All procedures were carried out isocratically. For organic acid determination, 25 mM phosphate buffer, pH 2.4, with methanol 1% was used as the eluent at a flow-rate of 1.0 mL/min. Separation was done at room temperature. Injection volume was 20 μ L and all standards and coffee samples were injected in triplicate. Quantitation of acids was performed at the wavelength of maximum absorbance for each analyte ($\lambda = 210$ nm) obtained from UV spectrophotometry spectra determination. Caffeine was separated with a mobile phase of 85% triethylamine phosphate buffer with 15% acetonitrile, pH 4.3, at a flow-rate of 1.0 mL/min, at room temperature. Injection volume was 20 μ L and detection was carried out at 270 nm.

2.5. Sensory evaluation

Sensory evaluation of brewed coffee samples analysed was performed according to ISO 6668 (1991) by a certified panel of tasters of Novadelta Company, Campo Maior, Portugal, who supplied all the 20 samples analysed (Table 2). The panel consisted of five judges trained by Novadelta. Samples were evaluated before and after roasting for physical and sensory attributes. Special concern was given to the sensory attributes of bitterness, sweetness and acidity in a scale of 1–4 (1—weak; 2—average; 3—strong; 4—very strong).

3. Results

3.1. Organic acid determination

3.1.1. SPE method

After identification of Strata-Sax as the most suitable sorbent for the analytes of interest, a method for extraction, separation and quantification of these compounds from brewed coffee samples was developed (Fig. 1a), aiming at quantification limits below 20 mg/L. All comparison experiments for the SPE method development were carried out with a set of standard solutions of citric, formic and quinic acids (concentrations from 150 to 300 mg/L).

3.1.1.1. Sample pre-treatment. The recovery of citric, formic and quinic acids was affected by pH. Recoveries were best with a 1:2 sample dilution step in 50 mM KH_2PO_4 buffer at pH 7.0. When using a mono-carboxylic acid pK_a range of 4–5 and a di-carboxylic acid pK_a range of 1.3–6.2 (SPE Reference Manual and Users Guide, Phenomenex), high recovery rates are expected at pH values of 6.0 or higher. In fact, good recovery rates were obtained at pH 7.0 where 99% ionization of analytes and retention is expected.

3.1.1.2. Conditioning step. Different conditioning solvents were tested: (1) Methanol and Milli-Q water; (2) methanol and 50 mM KH_2PO_4 buffer pH 7.0; and (3) NaOH 0.1 M.

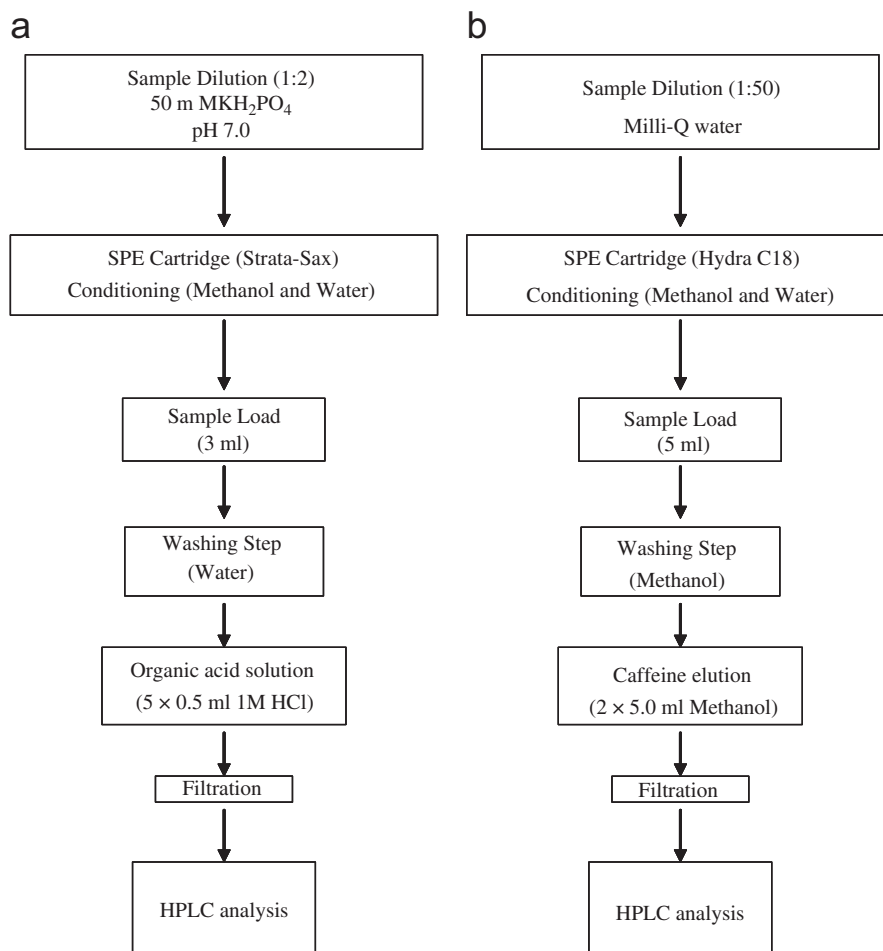


Fig. 1. Analytical procedure for the determination of (a) carboxylic acids and (b) caffeine in brewed coffee.

Table 2

Sensory evaluation (acidity, bitterness and sweetness) of brewed coffee from different types and geographic origins (1—weak; 2—average; 3—strong; 4—very strong)

Type	Sample identification	Roast	Acidity	Bitterness	Sweetness
Robusta	IA	Medium	2	2	2
	IB		2	3	2
	IC		3	3	2
	IDm	Light	1	3	2
	IDl		3	3	2
	IDd		1	4	1
	V	Medium	3	3	1
	C1		3	2	3
	C2		3	2	3
	Ci		1	3	1
Arabica	K	Light	3	2	3
	H1		3	2	3
	H2		3	2	3
	Z1l	Dark	3	3	1
	Z1d		3	3	3
	Z2		3	2	3
	Z3	Medium	3	2	2
	B		3	2	2
	T		3	3	2
	Co		2	3	1

Higher recoveries were seen when conditioning was done with methanol and Milli-Q water for the three tested acids.

3.1.1.3. Elution. Three elution methods were performed: (1) HCl 1 M; (2) HCl/methanol (1/99); and (3) H_2SO_4 0.1 M. The highest recovery rates were obtained with HCl 1 M as had been reported by Chinnici et al. (2005) with fruit juice.

3.1.1.4. Recoveries. To establish the efficiency of the organic acids SPE, the method was applied to a mixture of standard solutions. The results are shown in Table 3. This procedure was also performed on a mixture of organic acids added to brewed coffee (spiking). Table 3 shows the recoveries of these carboxylic acids after applying SPE method to brewed coffee.

3.1.2. HPLC

The proposed method allowed the identification and quantification of acetic, citric, formic, malic, piruvic, quinic and succinic acids in brewed coffee. These acids were identified by comparison of their retention times (in min) with those of standards and they were quantified by using an external standard calibration.

3.1.2.1. Limit of detection (LOD) and limit of quantitation (LOQ). The organic acid LODs were determined based on the standard deviation of the detector response (σ) and the slope (S) (Eq. (1); validation of analytical procedures: Methodology Q2B (ICH, 1996))

$$\text{LOD} = 3\sigma/S. \quad (1)$$

LOQ was expressed by

$$\text{LOQ} = 10\sigma/S. \quad (2)$$

The value of σ is calculated based on the experimental calibration curves being the standard deviation of y -intercepts of the regression line. During an extraction, only a fraction proportional to the extraction efficiency is available for analysis (Mitra and Brukh, 2003). S is the slope of the regression corrected for SPE recovery value (Table 4). HPLC calibration curves for the seven identified and quantified organic acids (acetic, citric, formic, malic, quinic, pyruvic and succinic) were corrected for SPE recovery values, and the slope of the corrected calibration curves was used to perform LOD and LOQ calculation (Table 4). LODs ranged from 0.4 mg/L for malic acid to 34 mg/L for acetic acid; QLs ranged from 1 mg/L for malic acid to 115 mg/L for acetic acid. LODs are close to other published results, even though the matrixes were different, e.g. fruit juices and nectars (Cunha et al., 2002; Chinnici et al., 2005) and honey (Suárez-Luque et al., 2002b).

3.1.2.2. Calibration curves. Calibration curves were determined for five different concentrations of standard solutions for each organic acid analysed. Each calibration standard was injected three times. The calibration test was performed in triplicate. Calibration curves were calculated applying the least-squares method to concentration versus peak area. Table 4 lists the parameters and correlation coefficients of the calibration plots. Each plot was linear over a wide interval from LOD to at least 200 mg/L for formic, pyruvic and quinic acid, 300 mg/L for citric acid, 900 mg/L for succinic acid, 1500 mg/L for acetic acid and 15 mg/L for malic acid. R^2 values were always above 0.9909 (except malic acid, 0.9890).

3.1.2.3. Repeatability. In order to express the precision under the same operating conditions over a short interval of time, repeatability of organic acids peak area was determined (Incedon and Lam, 2004) between 0.5% and 3.8% RSD. The results indicate that the present method can be applied to organic acids in brewed coffee. Intermediate precision expresses within-laboratory variation and is generally performed on different days using different analysts, equipment and sample preparations (Incedon and Lam, 2004). Calibration was repeated at the end of the study, and repeatability (again, 5 concentrations/3 replicates each) of organic acids peak area was between 0.2% and 1.6% RSD.

Table 3
Recoveries of standards added to coffee brew and of standard solutions after clean-up with SAX cartridges ($n = 3$)

Compound	Recoveries (%)	
	Spiked samples	Standard solutions
Acetic acid	112	39
Citric acid	95	122
Formic acid	24	76
Malic acid	80	59
Pyruvic acid	69	88
Quinic acid	63	103
Succinic acid	93	97

Table 5
Retention times for organic acids at different HPLC mobile phase pH values (in min)

	Mobile phase pH 2.4	Mobile phase pH 1.8
Acetic	3.388	3.310
Citric	3.983	4.450
Formic	2.134	2.270
Malic	2.286	2.620
Pyruvic	2.275	2.380
Quinic	1.983	1.950
Succinic	5.052	4.900

Table 4
Chromatographic parameters, concentration ranges and detection system of standard organic acids

Compound	T_r (min)	Capacity factor (k)	Range (mg/L)	Intercept	Slope	R^2 ($n = 3$)	LOD (mg/L)	LOQ (mg/L)
Acetic acid	3.388	1.19	50–1500	7114	324	0.9978	34	115
Citric acid	3.311	1.14	3–315	6922	559	0.9909	20	68
Formic acid	2.134	0.38	2.5–180	7315	1376	0.9980	7	23
Malic acid	2.286	0.48	1–12	−2762	22,736	0.9890	0.4	1
Pyruvic acid	2.275	0.47	2–230	−50,652	10,725	0.9957	9	31
Quinic acid	1.983	0.28	5–215	35,293	1102	0.9962	11	38
Succinic acid	5.052	2.27	60–900	11,658	356	0.9971	30	101

3.1.2.4. HPLC method development. Acetonitrile and methanol (1%) were tested as organic modifiers. Methanol gave better separation, which had already been reported by [Cawthray \(2003\)](#). The pH value of 1.8 was tried in order to improve separation. Retention times for all seven acids with pH values of 2.4 and 1.8 are shown in [Table 5](#). Malic and pyruvic acids had a better separation at pH values of 1.8 but separation of all other acids was almost the same ([Table 5](#)). Quantification of analytes was performed at the wavelength of maximum absorbance for each analyte ($\lambda = 210$ nm) obtained from UV spectrophotometry spectra determination.

3.1.2.5. Organic acid contents of brewed coffee. The organic acid contents of 20 brewed coffee samples are shown in [Table 6](#). Robusta showed a higher content of acetic, citric and succinic acids with light to medium roasts. These acids decreased in darker roast samples. Acetic, citric and succinic acids were the main acids found in brewed coffee, followed by formic and malic acids. Malic acid was present in our brewed coffee samples but at relatively low levels (always below 6 mg/L). In several samples, citric, formic, pyruvic and quinic were below LOQ in spite of recoveries of 60% or more in SPE (citric, 95%; pyruvic, 69%; quinic, 63%), which suggests that these acids are in very low concentrations in brewed coffee analysed. The organic acids detected in samples that are below the analytical LOQ were not quantified; the result is shown as <LOQ in [Table 6](#). If a specific acid was absent in a particular sample, we also referred to it as <LOD in [Table 6](#). Acid content varied among samples from the same origin and submitted to the same roasting conditions.

3.2. Caffeine determination

3.2.1. SPE method

Several eluants were tested for caffeine SPE (ethyl acetate, acetone, methanol) but the highest extraction efficiency was obtained with methanol, as had already been reported by [Buerge et al. \(2003\)](#). The SPE method ([Fig. 1b](#)) showed linearity for recovery rates up to 75 mg/L of caffeine. In this range, recovery rates for caffeine were between 96% and 100% (mean = 98.1, RSD = 2.5%). Chromatographic and method calibration parameters for caffeine determination are shown in [Table 7](#).

3.2.2. HPLC

Triethylamine is one of the components of the HPLC mobile phase for caffeine quantitation and shows a strong absorption at 270 nm (Srbek et al., 2005), although the triethylamine peak is well resolved from the caffeine peak and does not interfere in caffeine analysis. The detection and quantification limits were calculated the same way as for organic acid determination (Eqs. (1) and (2)). Repeatability (5 concentrations/3 replicates each) of caffeine peak area was between 0.5% and 2.3% RSD. Robusta coffee samples had higher caffeine content with a range from 314

Table 6
Organic acid content (mg/L) of the 20 brewed coffee samples analysed

Compound	Sample																			
	IA	IB	IC	IDm	IDI	IDd	V	C1	C2	Ci	K	H1	H2	Z1l	Z2	Z3	B	T	Co	
Acetic acid	<LOQ	296	588	530	662	531	1117	930	1309	823	265	<LOD	786	1221	730	2294	871	1068	<LOD	1379
Citric acid	nd	329	399	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOQ	<LOD	220	<LOD	495	<LOD	65	<LOD	<LOD	<LOQ	165	260
Formic acid	59	158	52	75	48	<LOQ	nd	<LOD	<LOQ	<LOQ	178	<LOQ	nd	nd	63	98	<LOD	<LOD	401	272
Malic acid	<LOQ	4	nd	3	3	3	nd	<LOQ	6	nd	nd	nd	nd	<LOQ	4	nd	nd	nd	3	3
Pyruvic acid	<LOQ	nd	<LOQ	nd	nd	nd	<LOD	<LOQ	<LOQ	<LOQ	<LOD	<LOQ	31	nd	nd	41	<LOD	<LOD	nd	<LOQ
Quinic acid	84	nd	<LOQ	<LOQ	<LOQ	<LOQ	nd	<LOD	<LOD	nd	<LOQ	<LOD	<LOD	<LOD	<LOD	nd	<LOD	nd	nd	nd
Succinic acid	585	1429	286	637	658	376	338	<LOQ	627	390	1402	524	449	nd	698	812	146	143	474	627
Total (mg/L)	728	2216	1295	1245	1371	910	1455	930	1942	1213	2065	524	1761	1221	1560	3245	887	1211	1043	2541

nd—not determined; LOQ—limit of quantitation; LOD—limit of detection.

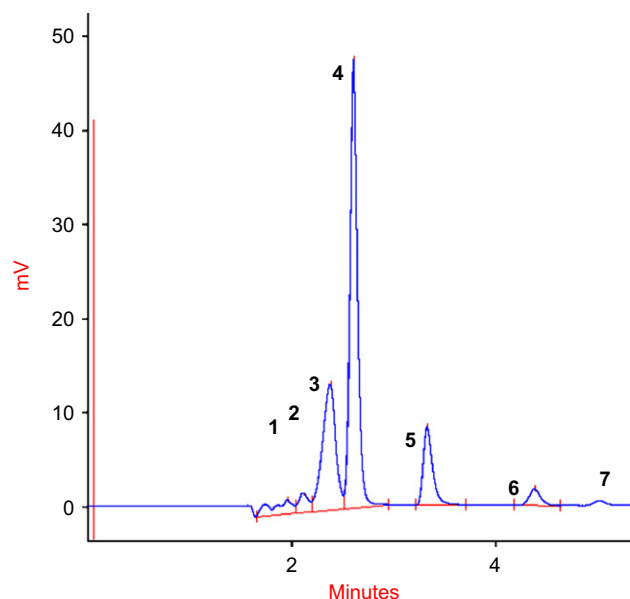


Fig. 2. Carboxylic acid separation at pH 1.8: (1) quinic acid, (2) formic acid, (3) pyruvic acid, (4) malic acid, (5) acetic acid, (6) citric acid, and (7) succinic acid.

Table 7
Chromatographic and method calibration parameters of caffeine determination

<i>HPLC separation</i>	
Caffeine retention time (min)	19.225
Capacity factor (<i>k</i>)	8.33
<i>HPLC calibration curve</i>	
Range (mg/L)	1–25
Slope	48177
Y-intercept	−1937.1
R^2 ($n = 3$)	0.9998
LOD (mg/L)	0.20
LOQ (mg/L)	0.67
<i>SPE calibration</i>	
Recovery (%) standards ($n = 3$)	98.1
Recovery (%) spiked sample ($n = 3$)	96

LOD—limit of detection; LOQ—limit of quantitation.

to 762 mg/L while arabica coffee showed lower levels of caffeine from 174 to 456 mg/L. Caffeine contents in brewed coffee from different types (robusta vs. arabica) and different geographic origins are shown in Table 1.

3.3. Sensory evaluation

Tasters reported an overall lower acidity and sweetness along with a higher bitterness for robusta coffees. The sensory evaluation shows a higher acidity in the case of Indian coffee when submitted to medium roast. The results are in accordance to the total acidity determined in the laboratory. The same was not observed when considering arabica coffees (for results of sensory evaluation, see Table 2).

4. Discussion

4.1. Organic acids in brewed coffee

The influence of parameters that affect the SPE process was studied in order to establish the optimal conditions for maximum recovery of organic acids and minimum extraction of interferences. An aqueous matrix (such as coffee brew) demands the selection of reversed-phase or ion-exchange sorbent depending on the ability of analytes for ionization (Hennion, 1999). Reversed-phase (C18) and ion-exchange cartridges had been tested in other works in progress in our laboratory (e.g. organic acid determination in culture media), and best recoveries were obtained with ion-exchange sorbents. Thus we decided to proceed with anion exchange for coffee sample clean-up as this sorbent is also applied by other authors for organic acid extraction from other food matrices (Cunha et al., 2002; Suárez-Luque et al., 2002b, Chinnici et al., 2005).

In this work, a SAX anion-exchange sorbent was selected, because coffee is a complex matrix and a very clean extract is required prior to HPLC analysis. SPE samples with high ionic strength must be diluted so we included a sample pre-treatment dilution step in the SPE methodology. According to Weiss (2004), ion-exclusion and cation exchange HPLC are characterized by high selectivity for aliphatic monocarboxylic acids with small numbers of carbon atoms, but sufficient separation cannot be obtained for the aliphatic open-chain and cyclic hydroxy acids that are also present in coffee. The author suggests that good separation is achieved by reversed-phase chromatography as has been used in this work. Nevertheless, low *k* values (Table 4) were obtained for some of the organic acids. When acetonitrile and methanol were tested as organic modifiers (at 1% concentration in mobile phase) the *k* values did not show significant differences. Further work took us to test a drop in pH value of the mobile phase to 1.8 which resulted in improved separation between malic and pyruvic acid peaks (Fig. 2). Retention times of all other peaks were almost the same (Table 5). This indicates that lower pH values could be essayed as long as the separation column is adequate to that in order to improve capacity factor and resolution.

We were unable to quantify lactic acid due to low recovery rates. With a pH value of 2.4 on the mobile phase, lactic acid oligomer formation can occur. Several peaks for this acid and its polymerization products were obtained in the determination of the acid in culture medium, which hindered its identification and/or quantitation. Lactic acid is subjected to intermolecular esterification even in dilute solutions, at low pH values, yielding the dimer lactoyllactic acid and also higher oligomers (Lucke and Göpferich, 2003; Ding and Schwendeman, 2004). Degradation of polymers occurs at higher pH and temperature values. Since the peak area for all other acids significantly decreases at pH values above 2.5 (Cawthray, 2003) this approach to lactic acid separation was not tried.

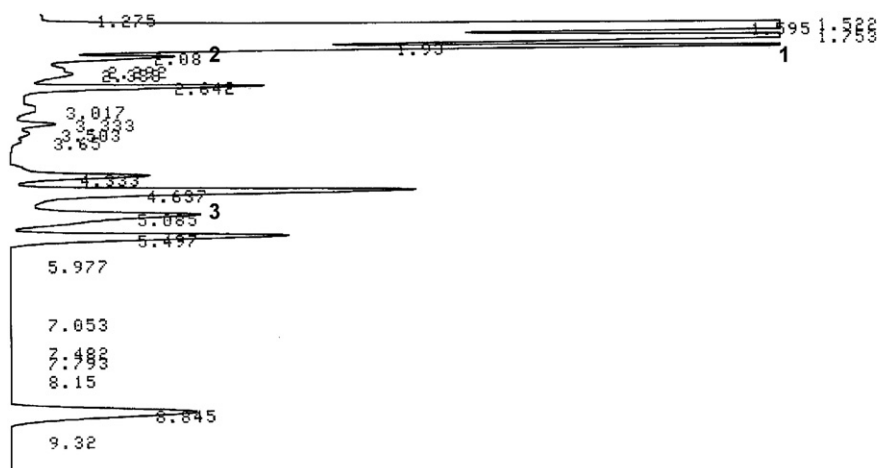


Fig. 3. India coffee (IA) after SPE: (1) quinic acid, (2) formic acid and (3) succinic acid.

The proposed method has the advantage of identifying and quantifying seven organic acids in brewed coffee with a sample preparation procedure similar to other authors (Suárez-Luque et al., 2002a, b; Chinnici et al., 2005) and a rapid chromatographic separation of 10 min using only one column (Fig. 3). It is generally accepted that changes in organic acid content, as a result of roasting, are responsible for the differences in beverage acidity that are observed when the same beans are roasted to different endpoints and then brewed identically (Clifford, 1989). Indian arabica coffee showed a higher total acidity with medium roast which agrees with results obtained by Balzer (1999) for both types of coffee (robusta and arabica) indicating the relevance of determining the best temperature and time of roast to enhance coffee sensory attributes. Although kinetic studies of acid production and decrease along roasting were not performed in this study, similar results were reported by Blanc (1977) with Colombian roasted coffee, by Ginz et al. (2000) and Verardo et al. (2002).

Acid content varied among samples from the same geographic origin (producer country) even when submitted to the same roasting conditions. For each origin, coffee can be cultivated in different soils, altitudes and seasons which may determine differences in the acidity of the green coffee and also after roasting.

4.2. Caffeine content

Although syringe barrel columns were used for caffeine SPE, the method can be adapted to SPE cartridges as long as proper sorbent mass is chosen improving the automation of the method. The mobile phase for caffeine determination contains a component that adsorbs at 270 nm, triethylamine, that appears in the chromatogram as a system peak (Srbek et al., 2005). This peak did not interfere with caffeine peak in our chromatographic system. The results obtained with the proposed method are in agreement with references. Robusta coffee contains higher levels of caffeine and chlorogenic acids, which are

partly responsible for bitterness and adstringency in coffee (Clifford, 1989). According to Campa et al. (2005), caffeine content (dry matter basis of green coffee bean) varies markedly between species and within species. *C. canephora* (robusta coffee), *C. brevipes* and *C. stenophylla* are the coffee species with the highest caffeine content, while *C. pseudozanguebariae* and *C. humblotiana* are almost caffeine-free.

The SPE and HPLC method proposed can be adapted for extraction and analysis of caffeine in other plant species. Tea (*Camellia sinensis* L.), cocoa (*Theobroma cacao*) and maté (*Ilex paraguariensis*) are other plant species of interest as regards caffeine content (Ashihara et al., 1996; Uno et al., 2005).

4.3. Sensory evaluation

Although sensory evaluation from the Novadelta panel of tasters agreed with the analytical results obtained for robusta coffee samples, the same did not hold for arabica coffee. The range of values for defining each attribute seems very narrow (only from 1 to 4). Also, the panel has never been calibrated with analytical evaluation of the brewed coffee, which might explain a higher discrepancy between tasters and the difficulty in relating our results with the evaluation performed by them.

5. Conclusions

As mentioned before, acidity of brewed coffee is an important feature that influences consumer reaction to the drink, and the acidity depends on the extent to which the coffee bean has been roasted. The determination of organic acids in beverage can indicate the best roasting conditions to apply to the green coffee or to a blend of green coffee in order to develop acidity and sweetness sensory attributes and improve sensory quality of commercialized roasted blends. The SPE method for organic acid extraction from brewed coffee can be used to GC–MS determination after

derivatization of the acids to the corresponding carboxylic acid esters although HPLC determination offers a less sensitive but less costly and faster option for this analysis. Determination of caffeine in brewed coffee samples through SPE and HPLC is a sensitive, clean and fast way to access caffeine content. The method can be adapted to other matrices such as soft drinks, pharmaceuticals and so on.

The application of SPE for extraction of the principal components from coffee samples provides a more environmentally sensitive alternative to liquid–liquid extraction, a relatively sensible and low-cost procedure compatible with many analytical methods. This work is included in a broader project to access important chemical composition profiles that, correlated with sensory analysis, should make it possible to build a database for identification of different coffee products and improve roasting process and develop new gourmet products. The final database will allow the most accurate calibration.

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